Reg No.:_____ Name:_____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Third Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: CHT201

Course Name: CHEMISTRY FOR PROCESS ENGINEERING

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks Marks

1	Construct an electrochemical cell for potentiometry & elucidate the function of each electrode involved in the measurement.	(3)
2	Explain half wave potential in polarography.	(3)
3	Discuss the basic principles of mass spectroscopy.	(3)
4	Describe the use of Atomic absorption spectroscopy for trace metal analysis.	(3)
5	Deduce the formula for distribution law in the case of association of solute in one of the phases.	(3)
6	Explain the terms (i) Consolute temperature (ii) Conjugate solution.	(3)
7	Discuss Langmuir's theory of adsorption.	(3)
8	What are emulsions? How is it prepared?	(3)
9	A bone taken from a garbage pile buried under a hillside had ^{14}C / ^{12}C ratio 0.477 times the ratio in a living plant or animal. What was the date when the animal was buried? Half life of carbon -14 is 5730 years.	(3)
10	What is meant by quantum yield in a photochemical reaction?	(3)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

11(a)	Describe constant current electrogravimetry with a neat diagram.	(7)
(b)	Explain the principle of Amperometry. Discuss Amperometric titrations when both the analyte & titrant is reducible.	(7)
12(a)	Write notes on (i) Ilkovic equation (ii) Half wave potential, and (iii) Migration Current.	(9)
(b)	Construct and explain the working principle of a biosensor.	(5)

Module 2

13(a) Give the principle, instrumentation & and any two applications of XPS. (7)

(b) Discuss the principle & instrumentation of Scanning Electron Microscope. Resolving power of Scanning Electron Microscope is higher than Optical Microscope. Justify. (7)

14(a) Explain the principle, instrumentation & any two disadvantages of AFM. (7)

(b) Explain the working principle and any three applications of AES. (7)

Module 3

15(a) Acetic acid was shaken with water and CCl_4 and the following concentration in gm. mol per litre were found in two layers. (6)

Water layer	5.02	7.98	10.70
CCl_4 layer	0.292	0.725	1.41

Assuming that acetic acid has single molecule in water show that it exists as double molecule in CCl_4

(b) Derive an equation for the rate constant of a first order reaction and deduce the expression for half-life of first order reaction. (8)

16(a) Derive an expression for primary salt effect. (7)

(b) Define upper and lower consolute temperature? With an example explain UCST and LCST both existing in same partially miscible liquid pair. (7)

Module 4

17(a) Derive an expression for Gibbs adsorption isotherm and explain Gibbs surface excess. (10)

(b) Discuss the functions of a protective colloid. (4)

18(a) Explain (i) Gold number (ii) Zeta potential (iii) CMC. (9)

(b) Write a note on surfactants. Give any two applications of it. (5)

Module 5

19(a) Explain Neutron activation analysis. Give any three applications. (7)

(b) Define (i) Mass Defect and (ii) Binding energy of a nucleus. How is Binding energy associated with the stability of a nucleus? (7)

20(a) What are the changes observed in the mass number & atomic number when an element decay by α and β emissions. (5)

(b) What is Nuclear reaction cross section? How is it related to reaction rate? (5)

(c) What is Chemiluminescence & bioluminescence? (4)

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